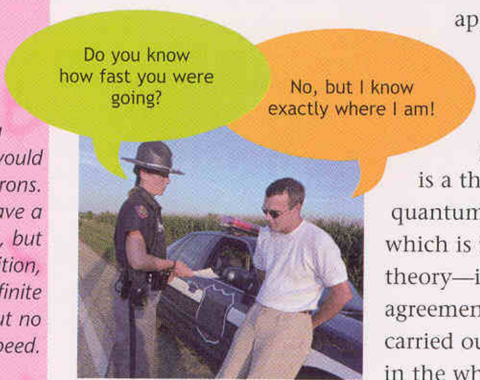
Werner Heisenberg (1901–76), one of the founding fathers of quantum theory, discovered the uncertainty principle, which says that it is not not possible to measure both speed and velocity on the quantum level. He clouded his reputation by being the most able scientist to stay in Germany after the Nazis came to power and working on the German atomic bomb project during World War II. However, his stature had been recognized by the award of the Nobel Prize in 1932.

life because it is so small, but it is very important for an electron. The uncertainty surrounding position and momentum is not simply a result of the difficulty of measuring the position and momentum of something as small as an electron; an electron itself does not "know" precisely both where it is and where it is going.

This package of ideas (uncertainty, probability, wave-particle duality, and the collapse of the wave function), all associated with the double-slit experiment, is the essence of the nature of the quantum world. Don't worry if it makes your head hurt—nobody understands quantum physics. What matters is that the equations associated with these ideas have many practical applications, which work whether you understand or not. We start our discussion of those applications with the jewel in the crown of quantum physics. This is a theory known as quantum electrodynamics, which is the most successful theory—in terms of its agreement with experiments carried out here on Earth—in the whole of science.



you can't know both

If Planck's constant were big enough, cars would act like electrons. They could have a definite speed, but no definite position, or a definite position, but no definite speed.

the jewel in the crown

The crowning achievement of quantum physics describes with great precision the way electrically charged particles interact with each other and with magnetic fields. This sounds esoteric, but it means that quantum electrodynamics, or QED, describes everything about matter that is not described by gravity. The everyday world is made up of atoms and molecules, which interact with each other through the electrons on the outer part of atoms. QED is able to account for all interactions involving electrons, and so QED is all that is needed to account for the everyday world. QED explains why the ocean is blue, how the explosions in an internal combustion engine take place, and how all chemical reactions take place. It is, in Richard Feynman's words, "the theory of light and matter" in all circumstances where gravity can be ignored. We only have to consider the other forces of nature (or "interactions," as physicists call them) when dealing with what goes on inside the nucleus of an atom—and, as we will see, even there QED provides the archetype on which other models are based. Without the example of QED, those models would have no foundations.

> **"All these 50 years of conscious brooding have brought me no nearer to the answer to the question, 'what are light quanta?' Nowadays every Tom, Dick, and Harry thinks he knows it, but he is mistaken."**
>
> Albert Einstein

at the heart of things

Quantum electro-dynamics pervades every aspect of the world we see around us. It explains the blue color of the ocean, and the way chemicals react with each other.